

Lasso Regression

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Regularisation ([Note 63](#)).
- **Leads to:** Elastic Net ([Note 69](#)).
- **Compare with:** Ridge regression ([Note 66](#)); L1 and L2 regularisation in neural networks ([Note 1026](#)).

1. Overview

Key point: Lasso is Ridge with the squares in the penalty replaced by absolute values. That one change lets it push coefficients to exactly 0, so it also removes useless **features** (G-772; input variables, the columns of the data table): automatic feature selection.

Lasso regression (G-1047) is the second regularised version of linear regression, also called **L1 regularisation** (G-1026). Like Ridge, it adds a penalty to the usual squared error, but the penalty uses the absolute size of each coefficient:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m |\beta_j|$$

	Ridge	Lasso
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Also called	L2 regularisation	L1 regularisation
Coefficients for large λ	close to 0, never exactly 0	exactly 0

LASSO stands for “least absolute shrinkage and selection operator” (Tibshirani 1996). As with Ridge, $\lambda \geq 0$: with $\lambda = 0$ Lasso is plain linear regression, a small λ may overfit, and a large one underfits. The intercept is not penalised.

This Note shows what Lasso does with one feature, with a flexible polynomial, and with the 10-feature diabetes data (442 **observations** (G-1374), that is, records or rows; the **target** (G-1949) we predict is disease progression), and why its coefficients can reach exactly 0.

2. One feature: the slope reaches exactly 0

Key point: As alpha grows, the Lasso slope falls in a straight line and stops at exactly 0.

Figure 1 fits Lasso to the 100-observation example of the Ridge Notes. scikit-learn calls the penalty strength **alpha**.

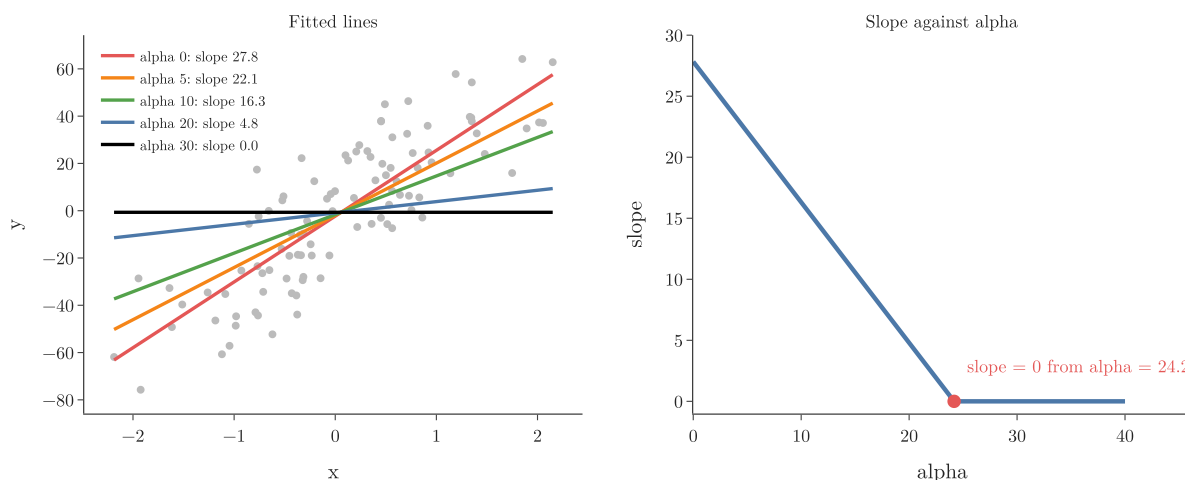


Figure 1: Lasso with one feature: fitted lines and the slope against alpha

alpha	Slope	Intercept
0 (linear regression)	27.83	-2.29
5	22.07	-1.96
10	16.31	-1.62
20	4.80	-0.95
25 and above	0	-0.67

From alpha 24.17 on, the slope is exactly 0. The line is then flat at the average of y (-0.67): the feature no longer plays any part. With Ridge, the slope only approached 0 (the Ridge maths Note).

Extra: scikit-learn's **Lasso** minimises $\frac{1}{2n} \sum (y_i - \hat{y}_i)^2 + \alpha \sum |\beta_j|$ (scikit-learn docs, **Lasso**): it averages the squared error and halves it. So its **alpha** is on a different scale from the λ of the formula above, and from **Ridge's alpha**. The values of alpha for Ridge and Lasso cannot be compared directly.

3. A flexible model: Lasso picks the right terms

Key point: On a degree-16 polynomial, a moderate alpha set 14 of the 16 coefficients to 0 and kept exactly x and x^2 , the true shape of the data.

Figure 2 fits a degree-16 polynomial to 100 points made from $y = 0.7x^2 - 2x + 3$ plus noise. The 16 power columns (x to x^{16}) are standardised first.

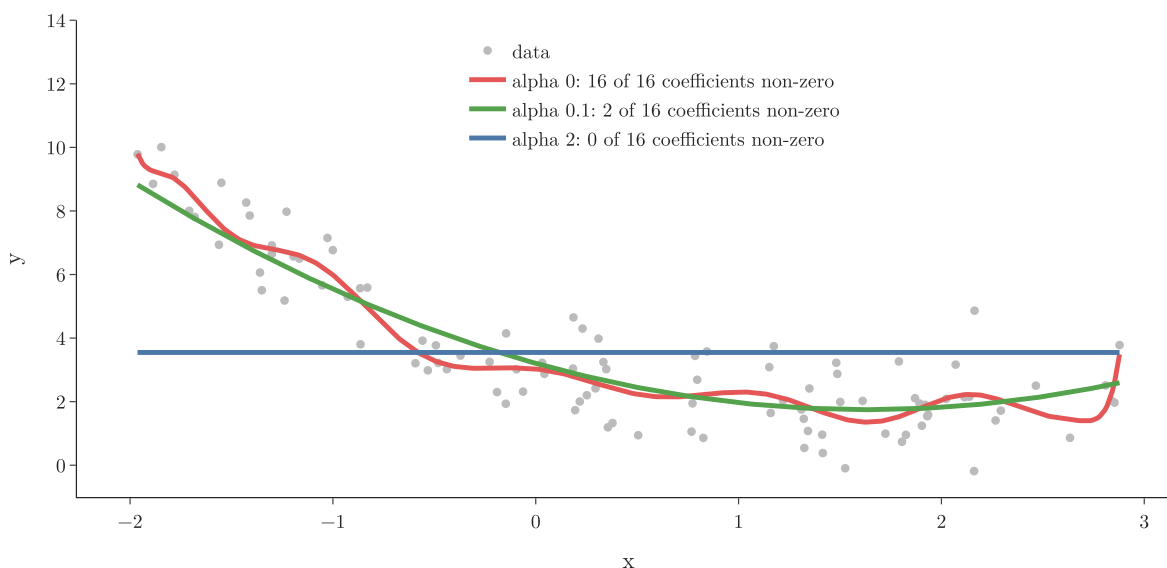


Figure 2: Degree-16 polynomial with Lasso for three values of alpha

- **alpha 0** (red): all 16 coefficients are used, and the curve wiggles to follow the noise, swinging up at the right edge: overfitting.
- **alpha 0.1** (green): only 2 coefficients are non-zero, those of x and x^2 . The curve is a smooth parabola, the true shape.
- **alpha 2** (blue): every coefficient is 0, and the model predicts the average everywhere: underfitting.

alpha	Non-zero coefficients
0.01	x, x^2, x^3, x^{16}
0.1	x, x^2
1	x

Lasso found which of the 16 power features matter without being told.

4. Feature selection

Key point: Coefficients that reach 0 remove their features from the model. So Lasso selects features while it trains.

A coefficient of exactly 0 means that feature has no effect on the predictions, so its column can be dropped. A model in which many coefficients are exactly 0 is a **sparse model** (G-1844). Zero coefficients make Lasso a tool for **feature selection** (G-768; see the [feature engineering Note](#)), useful when there are many features and some of them do not matter. Think of packing a suitcase with a strict weight limit: the items you barely need are left out entirely.

Figure 3 trains Lasso on the diabetes data (test size 0.2, random state 2).

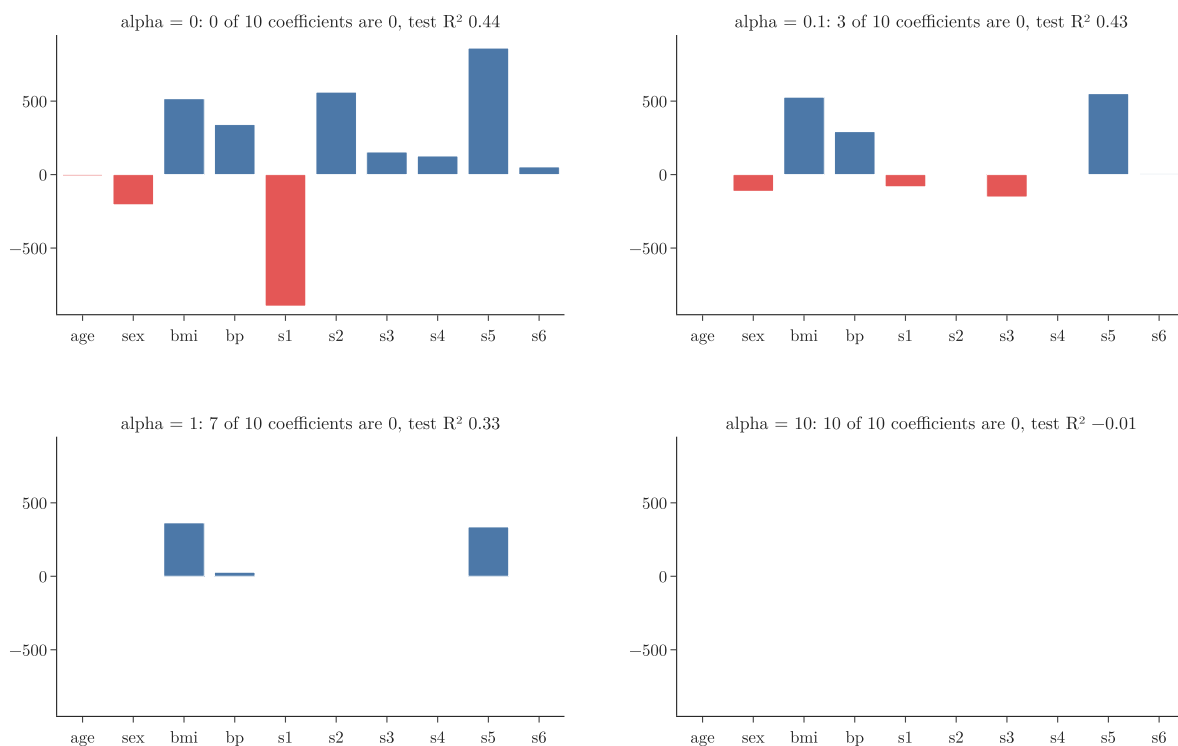


Figure 3: Diabetes coefficients for α 0, 0.1, 1 and 10, on the same scale

α	Coefficients at 0	Features kept	Test R^2
0	0	all 10	0.44
0.1	3	7 (age, s2, s4 dropped)	0.43
1	7	bmi, bp, s5	0.33
10	10	none	-0.01

At α 0.1, Lasso drops three features and loses almost no accuracy (0.43 against 0.44). At α 1, Lasso keeps only bmi, bp and s5, the three most useful features here. At α 10, Lasso removes everything: underfitting.

Ridge cannot do this. Ridge shrinks an unhelpful coefficient to, say, 0.2, but the feature stays in the model.

A test with useless features. Suppose we predict a mouse’s size from its weight, its diet, its astrological sign and the airspeed of a swallow. The last two are useless. Ridge would shrink their coefficients but keep them in the equation; Lasso can remove them (StatQuest, “Regularization Part 2”). Figure 4 runs this test on the diabetes data. We add 10 columns of pure random noise, on the same scale as the 10 real features, and fit both models.

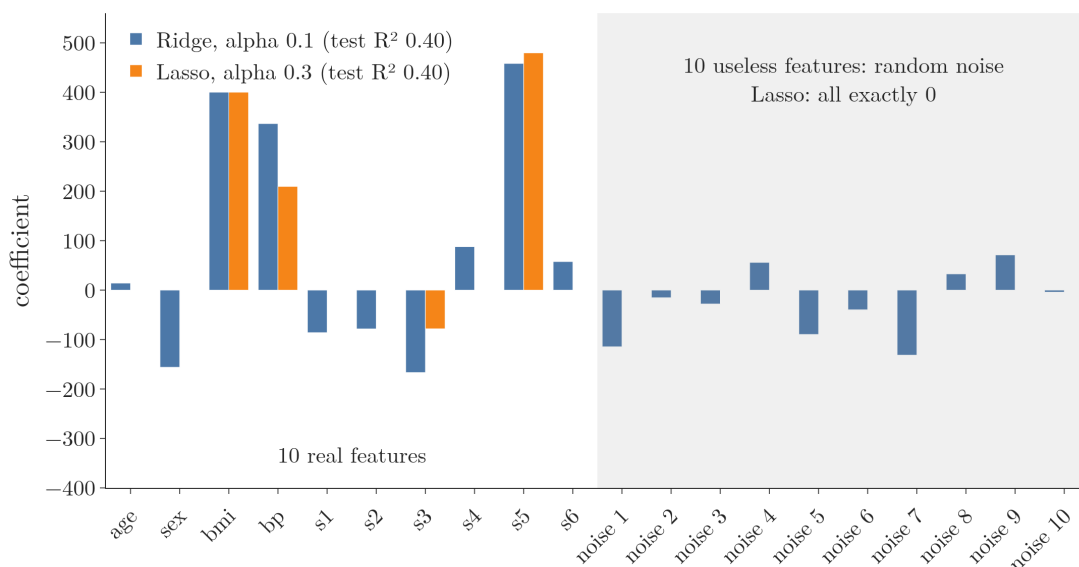


Figure 4: Coefficients on the diabetes split with 10 extra columns of random noise (shaded). Ridge with alpha 0.1 (blue) gives every noise column a non-zero coefficient. Lasso with alpha 0.3 (orange) sets all 10 noise coefficients to exactly 0 and keeps bmi, bp, s3 and s5. Both score test R^2 0.40.

- **Ridge** (alpha 0.1) gives all 10 noise columns a coefficient, some as large as -131 .
- **Lasso** (alpha 0.3) sets all 10 to exactly 0 and keeps four real features: bmi, bp, s3 and s5.
- Both reach test R^2 0.40, against 0.38 for plain linear regression on the 20 columns.

The result does not depend on one lucky draw of noise: over 100 different draws, Lasso zeroes 9.65 of the 10 noise columns on average, and Ridge none (script `useless.py`).

Extra: Why call bmi, bp and s5 the most useful three? Among all 120 possible sets of three features, these three give plain linear regression the best cross-validated R^2 (0.485).

Extra: Feature selection is why Lasso is often preferred when the data has many columns, some of which are probably irrelevant: Lasso tends to do best when only a few features really matter (ISL §6.2.2). Fewer features also means a simpler model that is easier to explain (ISL §6.2.2).

5. How the coefficients change

Key point: The coefficients hit 0 one at a time. The size of a coefficient does not decide when it goes: the large s2 goes second, while bmi and s5 survive longest.

Figure 5 follows each coefficient as alpha grows on a log scale.

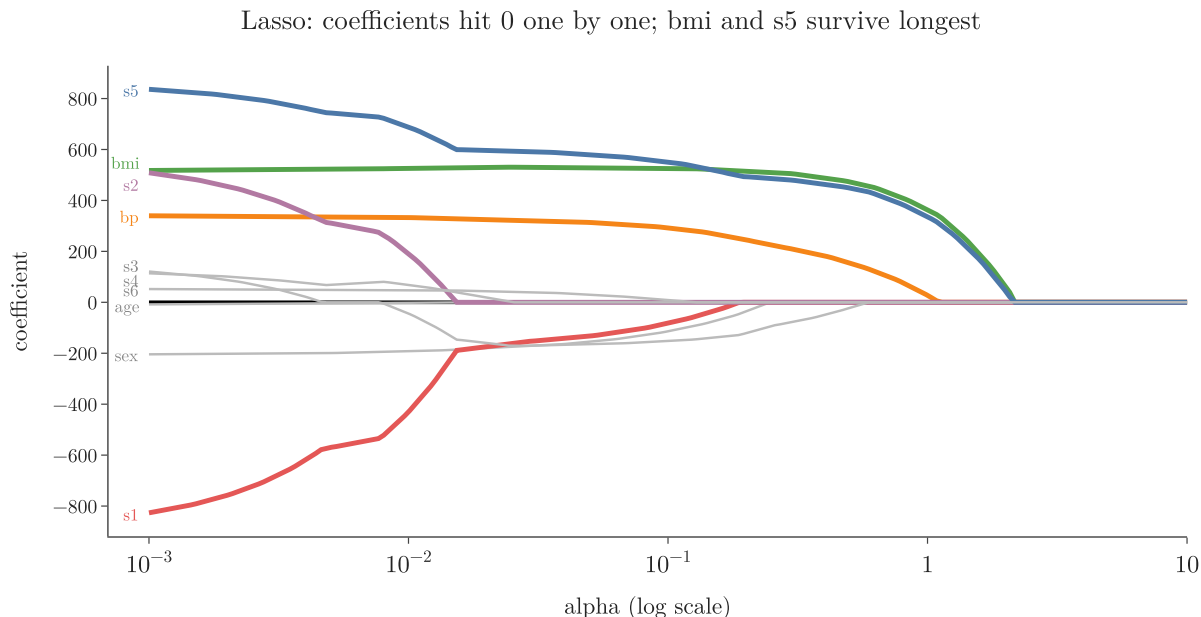


Figure 5: Lasso coefficient paths on the diabetes data

Feature	Coefficient at alpha 0	Exactly 0 from alpha about
age	-9	0.012
s2	561	0.015
s4	127	0.026
s1	-896	0.20
bp	341	1.1
bmi, s5	517, 861	2.2

- As with Ridge, the very large coefficients (s1, s2, s5) are cut back hard at first. The cut barely changes the test R^2 (0.440 at alpha 0, 0.441 at alpha 0.01, 0.433 at alpha 0.1).
- Then the coefficients drop to 0 one after another. bmi and s5 are the last to go.
- s1 and s2 had huge coefficients (-896 and +561) because they are strongly correlated with each other ($r = 0.90$): such a pair can get large opposite coefficients that partly cancel out (ESL §3.4.1). Lasso drops them early anyway: size alone does not protect a coefficient.

6. Bias and variance

Key point: As with Ridge, a larger alpha raises bias and lowers variance. The best alpha is in between.

We use the same method as the [Ridge key points Note](#): made-up data from the known curve $y = 0.7x^2 - 2x + 3$ plus noise (standard deviation 2), 20 fixed training observations, a

degree-16 polynomial, and 200 fresh noise draws. Bias² and variance are measured against the true curve at 19 test points between the training ones.

alpha	Bias ²	Variance	Expected test error
0.001	0.009	1.44	5.45
0.01	0.009	0.95	4.96
0.1	0.04	0.62	4.66
0.3	0.34	0.54	4.88
1	2.52	0.38	6.90
3	5.20	0.19	9.40

Figure 6 shows the experiment. At a tiny alpha the 200 fitted curves fan out, most of all at the edges: high variance. At alpha 3 every fit is a flat line at the average, the same for every draw: almost no variance, but far from the true curve. At alpha 0.1 the fits hug the true parabola.

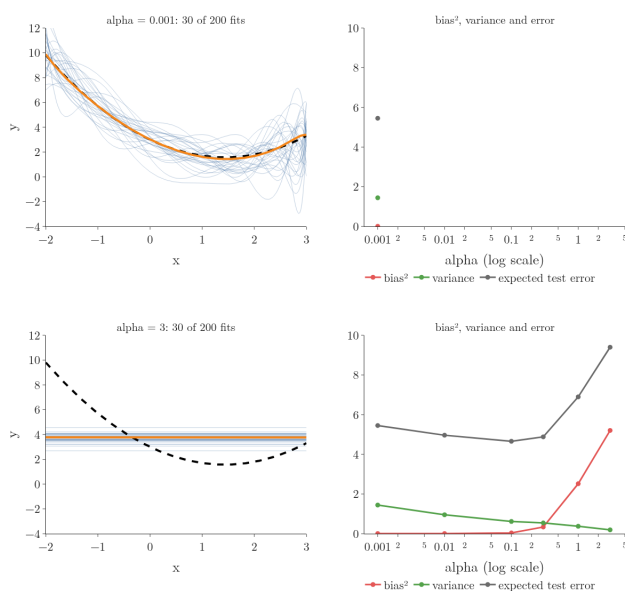


Figure 6: Bias and variance of Lasso on a degree-16 polynomial. Left: 30 of the 200 fitted curves (blue), their average (orange) and the true curve (dashed) for each alpha. Right: bias², variance and expected test error, built up alpha by alpha; the error is lowest at alpha 0.1.

Small alpha: high variance (**overfitting**, G-1429). Large alpha: high bias (**underfitting**, G-2035). Here alpha 0.1, the same value that kept only x and x^2 in Section 3, has the lowest total error.

Extra: Expected test error = bias² + variance + noise (ISL §2.2.2, eq. 2.7); the noise part is $2^2 = 4$, the floor no model can beat.

7. Why coefficients reach exactly 0

Key point: The penalty $\lambda|m|$ makes a sharp corner in the loss at $m = 0$. Once λ is large enough, that corner is the lowest point, and it stays there however large λ gets.

Take the one-feature example again and hold the intercept at -2.29 , so the loss depends only on the slope:

$$L(m) = \sum_{i=1}^n (y_i - mx_i + 2.29)^2 + \lambda|m|$$

Figure 7 draws this curve while λ grows from 0 to 8,000.

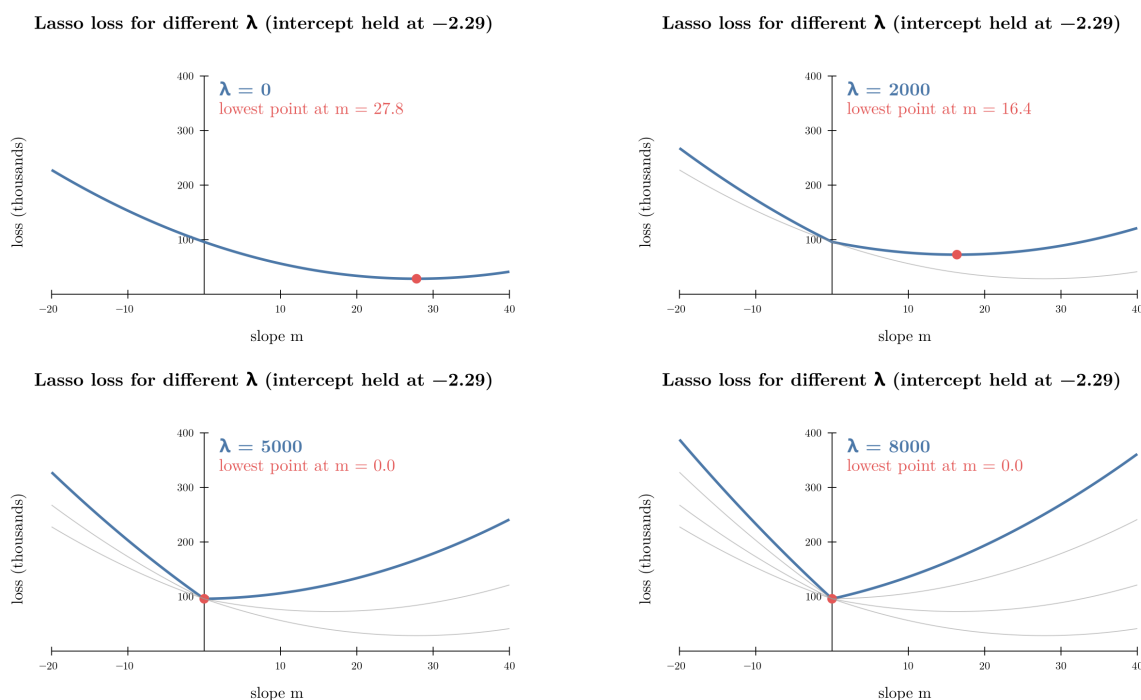


Figure 7: The Lasso loss against the slope for $\lambda = 0, 2000, 5000$ and 8000

λ	Slope at the lowest point
0	27.8
2,000	16.4
5,000	0
8,000	0

- The penalty $\lambda|m|$ is a V shape: it is 0 at $m = 0$ and grows at the same rate on both sides. Added to the smooth bowl, it gives the curve a sharp corner at $m = 0$.
- As λ grows, the lowest point moves left towards 0, as with Ridge.

- Once λ passes about 4,850 here, the corner itself is the lowest point. From then on the answer is exactly $m = 0$, and a larger λ only makes the corner steeper.

The Ridge penalty λm^2 is smooth and flat at $m = 0$, with no corner, so its lowest point only approaches 0 (the Ridge key points Note). The next Note derives this precisely.

Figure 8 puts the two loss curves next to each other, with the same λ in both panels. Watch the red dots. On the left, the Ridge curve stays a smooth bowl; its lowest point glides towards 0 and is still at 0.30 when $\lambda = 8,000$. On the right, the Lasso curve grows a corner at $m = 0$, and from $\lambda = 5,000$ the lowest point sits exactly on that corner.

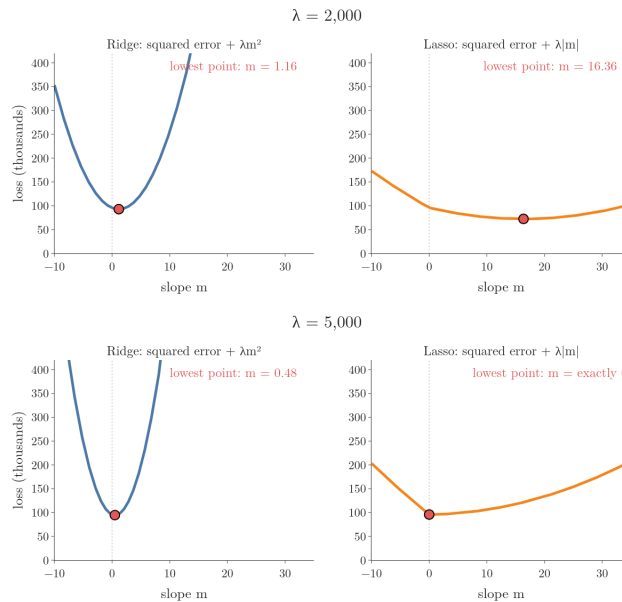


Figure 8: The Ridge loss (left) and the Lasso loss (right) against the slope, for the same λ from 0 to 8,000. The red dot marks the lowest point: Ridge’s glides towards 0 without reaching it; Lasso’s reaches exactly 0 at $\lambda = 5,000$ and stays. Idea after StatQuest, “Ridge vs Lasso Regression, Visualized!!!”

Extra: Why do the two penalties behave so differently near 0? Look at how hard each penalty pushes, which is its slope. The slope of λm^2 is $2\lambda m$: at $m = 0.1$ it is only 0.2λ , and it fades to 0 as m approaches 0, so Ridge stops pushing. The slope of $\lambda|m|$ is λ for every positive m : it pushes with the same force all the way to 0.

8. Ridge or Lasso?

Key point: Many features, some probably useless: Lasso. Most features useful, especially correlated ones: Ridge.

	Ridge	Lasso
Penalty	squares, $\lambda \sum \beta_j^2$	absolute values, $\lambda \sum \beta_j $
Coefficients	shrink, never exactly 0	shrink, many become exactly 0
Feature selection	no	yes
Best when	most features matter	only some features matter
Closed-form formula	yes	no (solved step by step)

The “best when” row follows ISL §6.2.2: Lasso does a little better when many features are useless, Ridge when most are useful (also StatQuest, “Regularization Part 2”). Lasso has no closed-form formula because the absolute values make the answer non-linear in y (ESL §3.4.2). With strongly correlated features, Lasso tends to keep only one of the group, and Ridge has been seen to predict better (Zou and Hastie 2005, §1).

Figure 9 puts the two side by side on the diabetes split, both with alpha 0.1. Ridge keeps all ten coefficients non-zero; Lasso sets age, s2 and s4 to exactly 0 and pays only 0.02 of test R^2 for it.

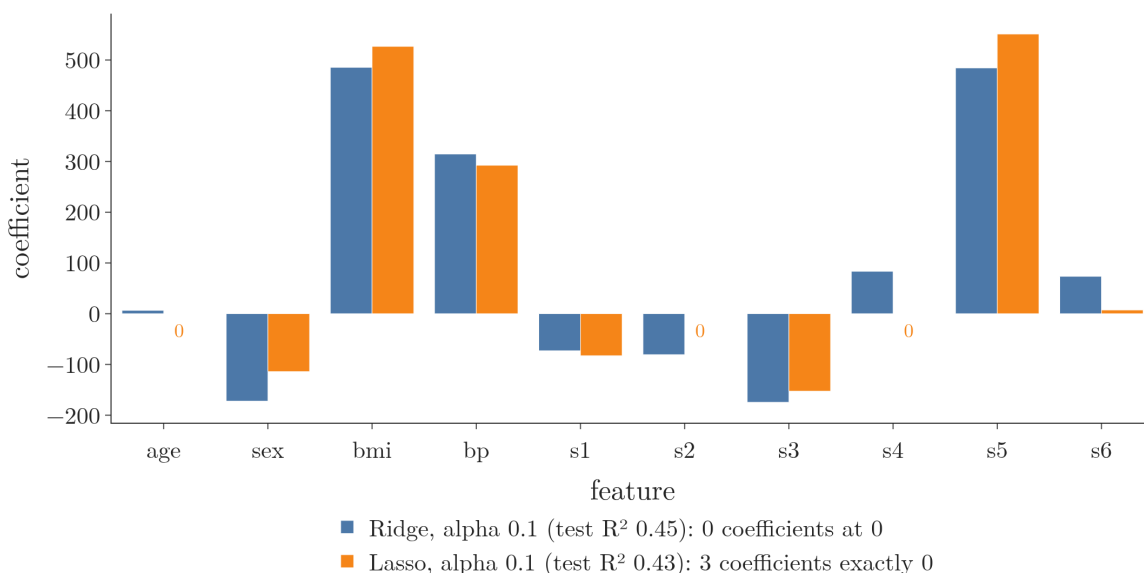


Figure 9: Coefficients on the diabetes split with alpha 0.1. Ridge (blue, test R^2 0.45) keeps all ten features; Lasso (orange, test R^2 0.43) sets age, s2 and s4 exactly to 0.

Both shrink the largest coefficients, raise bias, lower variance and are tuned through λ . The difference that matters in practice is that Lasso can remove features. **Elastic Net** (G-668; a later Note) combines both penalties.

Python: Lasso in scikit-learn.
`from sklearn.linear_model import Lasso`

```
lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)
lasso.coef_           # several exactly 0
lasso.score(X_test, y_test) # R2 0.43 on the diabetes split
As with Ridge, features should be on the same scale first, for example with
make_pipeline(StandardScaler(), Lasso(alpha=0.1)).
```

9. Summary

- Lasso adds $\lambda \sum |\beta_j|$ to the squared error: L1 regularisation.
- As λ grows, coefficients shrink and then become exactly 0, one by one.
- Zero coefficients remove features: Lasso performs feature selection (diabetes: alpha 1 keeps bmi, bp and s5; the polynomial: alpha 0.1 keeps x and x^2).
- A larger λ means more bias and less variance; too large removes every feature.
- The absolute value makes a corner at 0 in the loss, which is why the answer can sit exactly at 0.

10. Sources

Built from

- CampusX, “Lasso Regression | Intuition and Code Sample | Regularized Linear Models”, YouTube, <https://www.youtube.com/watch?v=HLF4bFbBgwk>
- StatQuest with Josh Starmer, “Regularization Part 2: Lasso (L1) Regression”, YouTube, <https://www.youtube.com/watch?v=NGf0voTMIcs>. The useless-variables example (Section 4) and the Ridge-or-Lasso guidance (Section 8).
- StatQuest with Josh Starmer, “Ridge vs Lasso Regression, Visualized!!!”, YouTube, https://www.youtube.com/watch?v=Xm2C_gTAI8c. The side-by-side loss curves (Figure 8), redrawn on our own data.

Other references

- **Tibshirani 1996:** Tibshirani, R. “Regression Shrinkage and Selection via the Lasso.” *Journal of the Royal Statistical Society B* 58(1), 267–288, 1996. Section 1.
- **ISL:** James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 2.2.2, p. 34; Section 6.2.2, pp. 242, 246.
- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Sections 3.4.1 (p. 63) and 3.4.2 (p. 68).
- **Zou and Hastie 2005:** Zou, H. and Hastie, T. “Regularization and Variable Selection via the Elastic Net.” *Journal of the Royal Statistical Society B* 67(2), 301–320, 2005. Section 1.
- **scikit-learn docs:** `sklearn.linear_model.Lasso`, scikit-learn 1.9.

11. Key terms

Term	Meaning
Lasso regression	Linear regression with a penalty on the sum of absolute coefficients
L1 regularisation	Another name for the absolute-value penalty used by Lasso
Sparse model	A model in which many coefficients are exactly 0